Fourteen Caption Users

AP Statistics · Unit 3 · Topic 3.2 · B: 2026-11-23 · E: 2026-12-21 · TEACHER KEY

CED TETHER

- **Skill 3.B** — Determine parameters for probability distributions.
- **Skill 3.C** — Describe probability distributions.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- **LO UNC-3.K** — Determine parameters of a sampling distribution for sample proportions. [Skill 3.B]
- **EK UNC-3.K.1** — For independent samples of a categorical variable from a population with proportion p, the sampling distribution of the sample proportion has mean μ = p and a standard deviation σ = sqrt(p(1-p)/n).
- **EK UNC-3.K.2** — If sampling without replacement, the standard deviation of the sample proportion is smaller than the formula above. If the sample size is less than 10% of the population size, the difference is negligible.
- **LO UNC-3.L** — Determine whether a sampling distribution for a sample proportion can be described as approximately normal. [Skill 3.C]
- **EK UNC-3.L.1** — For a categorical variable, the sampling distribution of the sample proportion will have an approximate normal distribution provided the sample size is large enough: np $\geq$ 10 and n(1-p) $\geq$ 10.
- **LO UNC-3.M** — Interpret probabilities and parameters for a sampling distribution for a sample proportion. [Skill 4.B]
- **EK UNC-3.M.1** — Probabilities and parameters for a sampling distribution for a sample proportion should be interpreted using appropriate units and within the context of a specific population.

Essential question: How do sampling, spread, and shape conditions limit what a surprising sample proportion can show?

DOK-3 skill: 4.B · **FRQ pattern:** `failed-large-counts-normal-tail-vs-valid-binomial-and-bias`

DOK rationale: Part (c) separates independence, shape, and unbiasedness to evaluate two overclaims and use a supplied probability to bound the conclusion supported by a random sample.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) DOK: 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Ask students to mark which sentence in each claim is about shape, center, or evidence.
- Begin the lesson video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and press the distinct roles of 10 percent and Large Counts.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose whose reasoning is closer and name one condition before the lesson.
- Complete the follow-along about the center, spread, shape, and probabilities for sample proportions.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Which condition supports treating the 100 selections as approximately independent?
- Why does $np = 8$ address shape rather than the estimator's long-run center?
- What values of X belong to $\widehat{p} \geq 0.14$?
- How do 0.0135 and about 0.028 change the strength, but not the direction, of the evidence?

Adult role

Solo teacher. Circulate 5 pairs. Require students to attach each condition to its job and to distinguish evidence against 0.08 from proof of a changed population rate.

[WARN] WATCH FOR

- Applying an $n > 30$ rule carried over from sample means.
- Calling a nonnormal sampling distribution biased.
- Calculating exactly 14 rather than 14 or more caption users.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Uses the passing 10 percent condition and failed np=8 normal-shape condition to distinguish the two model requirements.
- **judgment-2** (required) — Uses supplied tail near 0.0282 as unusual evidence under p=0.08, without claiming proof of an increase.
- **judgment-3** (required) — Explains that SRS unbiasedness concerns center, whereas normality concerns shape, and rejects both overclaims.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

Common mistakes

- Checking only $n > 30$ instead of both expected counts
- Using $P(X = 14)$ rather than the upper-tail event $P(X \geq 14)$
- Saying the 10 percent condition determines normal shape rather than approximate independence and spread
- Equating a skewed sampling distribution with a biased estimator

[STAR] TODAY'S PROBLEM — annotated

Suppose a video platform has 5,000 active accounts and says that $p = 0.08$ of them regularly use captions. An SRS of 100 accounts, selected without replacement, contains 14 regular caption users, so $\widehat{p} = 0.14$.

Jules: "Because $n = 100 > 30$, $\widehat{p}$ is approximately normal. I get $P(\widehat{p} \geq 0.14) = 0.0135$, which proves caption use increased."

Nora: "Because $np = 8 < 10$, a normal model is not justified. Therefore $\widehat{p}$ is biased and this sample tells us nothing."

A binomial calculator gives $P(X \geq 14) = 0.0282$ for $X \sim \text{Bin}(100, 0.08)$. This closely approximates sampling from 5,000 accounts.

Caption-use sample

Source	Accounts	Users	Rate
Population	5,000	400	0.08
SRS	100	14	0.14

FIRST TAKE (5 min)

Does finding 14 caption users instead of the expected 8 prove that use increased? Give one reason.
Key: Not graded. Each student has one correct observation and one overreach.

Rules callout "CHECK INDEPENDENCE, THEN SHAPE (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Identify p and verify the sample proportion $\widehat{p} = 14/100$.

Key: The parameter is p, the proportion of all 5,000 active accounts that regularly use captions. The statistic is the sample proportion $\widehat{p} = 14/100 = 0.14$. Thus $N = 5{,}000$ and $n = 100$.

DOK 2 **(b)** Find the mean and standard deviation of the sampling distribution of $\widehat{p}$. Check the 10 percent condition and both Large Counts values.

Key: The sampling distribution has mean $\mu_{\widehat{p}} = p = 0.08$ and standard deviation $\sigma_{\widehat{p}} = \sqrt{0.08(0.92)/100} = 0.027129\ldots \approx 0.0271$. The 10 percent condition holds because $100 < 0.10(5{,}000) = 500$, so dependence from sampling without replacement is negligible. Large Counts does not fully hold: $np = 100(0.08) = 8 < 10$, while $n(1-p) = 100(0.92) = 92 \geq 10$.

DOK 3 **(c)** In 3–4 sentences, explain what each student gets right and wrong. Use the condition checks and the supplied probability 0.0282. State what the sample suggests about caption use and why a failed normal-shape check does not make the estimate biased.

Key: Neither claim is fully correct: the sample is random and $100 < 500$, but $np = 8 < 10$, so Nora correctly rejects the normal approximation. The supplied binomial tail, about 0.0282, makes 14 or more unusual under $p = 0.08$, giving evidence worth investigating rather than proof of an increase. The SRS produces an unbiased sample proportion; the failed condition concerns shape, not its long-run center. Jules overstates the evidence and Nora wrongly discards a useful estimate.

EXIT

One thing the video changed about my first take: ______________________________